

Intro to Computational Intelligence:

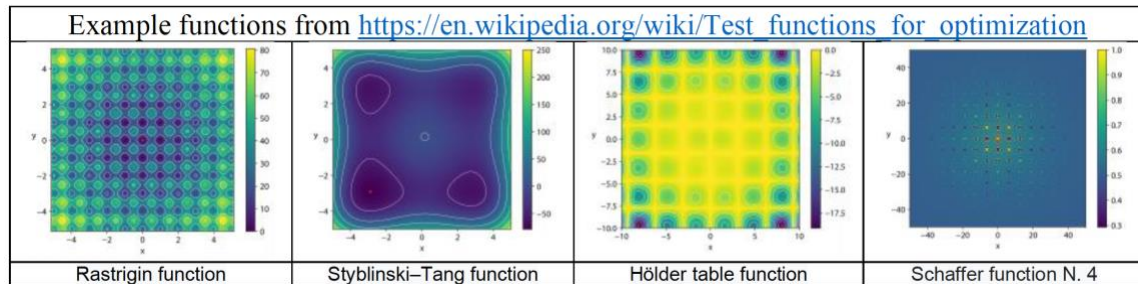
Project 1 – Evolutionary Algorithms

Project Description

In this project (P1), your goal is to design, implement, and analyze evolutionary algorithms, with a specific focus on Genetic Algorithms (GAs). The purpose is not only to code up a GA, but also to systematically experiment with its design choices and analyze its performance.

Tasks for all students:

- Choose three single-objective unconstrained optimization functions of different complexity from https://en.wikipedia.org/wiki/Test_functions_for_optimization and explain why you picked those functions.



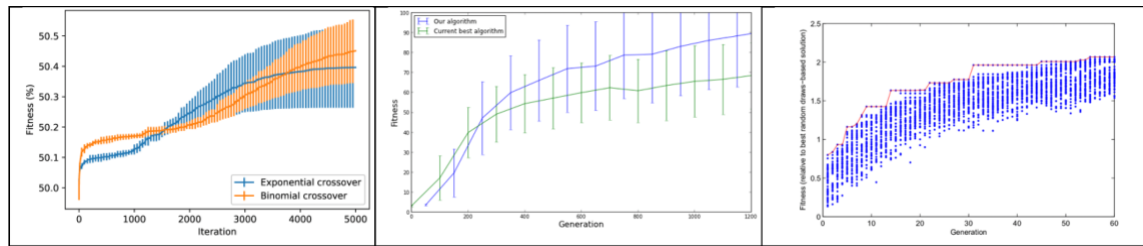
- Implement a GA and systematically vary two: (i) initialization methods (e.g., random vs exemplar), (ii) selection mechanisms (e.g., proportional, tournament, rank based), (iii) crossover operators (e.g., two-point, mask), (iv) mutation operators, and (v) termination criteria. In total, this yields $2^5=32$ different GA variants.
- Construct a GA vs fitness function performance table. Report average (the 0.9 below) and standard deviation (the 0.2 number in ()'s) of performance across experiments.

	Fx1	Fx2	...
GA1	0.9 (0.2)	...	
GA2			
...			

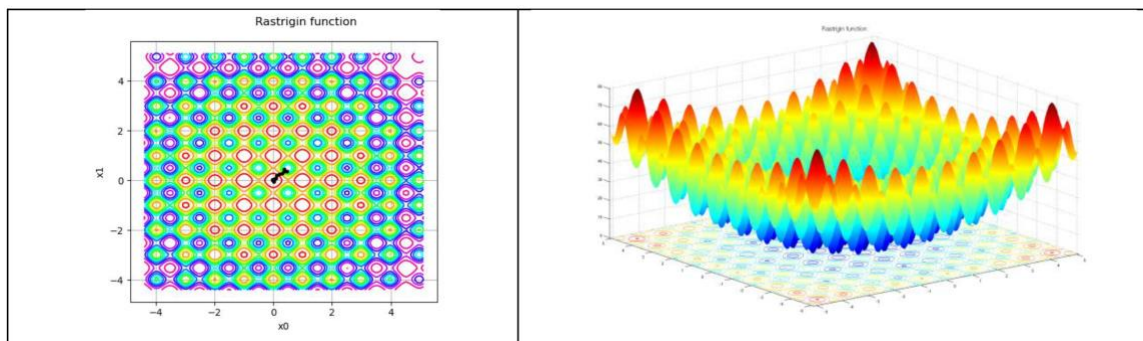
- For one chosen function, construct a comparative table of multiple performance measures (average, min, max fitness, std deviation, etc.).

	Fitness	Min Fitness	...
GA1	0.9 (0.2)		
GA2			
...			

- Generate plots of fitness vs generations for selected GA variants. Compare trends such as steady learning, premature convergence, or random-like behavior.



- Create 2D or 3D contour plots of functions with solution populations shown per generation. Animate solution movement to illustrate GA dynamics.



Graduate students (in addition to the above):

Pick 3 from the following 5

1. Extend your analysis to higher dimensions (e.g., $D=10, 100, 1000$) and compare to $D=2$.
2. Explore an advanced operator, e.g., multi-parent crossover such as UNDX/SPX/PCX, or advanced mutation strategies such as Gaussian.
3. Conduct systematic experiments on mutation vs crossover balance. Analyze trade-offs between exploration and exploitation.
4. Explore a strategy like cultural evolution, differential evolution, or co-evolution.
5. Apply your GA to a real dataset / problem. Examples: training a classifier, learning rules, learning reinforcement learning policies, optimizing the structure (network architecture search?) or weights of a neural network, your graduate research, etc.

NOTE, undergraduates can earn up to 6% extra credit (2% each) by completing graduate-level extensions. Don't want y'all to miss out on the fun!

Report Requirements

Your final report must follow these guidelines:

1. Cover Slide: Title, name, student ID, date.
2. Introduction (5–10 pages): In your own words (no genAI, no online, no book), explain (i) what an evolutionary algorithm is, (ii) their connection to nature, (iii) their general

structure and parts and how do they work, and (iv) what specific operators/strategies did you implement in this project. Demonstrate mastery of the material.

3. Experiments and Results (10–20 pages): Present experiments clearly with reproducibility in mind. Label figures, axes, tables, and explain your design choices. Avoid result dumps—summarize findings, highlight expected and unexpected results, and discuss insights.

4. Summary and Future Work (3–5 pages): Discuss overall findings, what you learned, and potential next steps.

Code Submission

Submit your code as a single ZIP file (under 50MB, no data). Ensure reproducibility and clarity in code organization.

Rules and AI/GenAI/LLM Use Policy

You must work individually. Collaboration is not allowed. The ONLY exception is AI. You **may** use tools such as ChatGPT for coding up helper routines (e.g., plotting, data loading, formatting), optimizing your code (e.g., a single generation taking 2 minutes vs 20 minutes), and for polishing your report (e.g., helping with grammar, formatting, organization).

However, all core GA algorithm code must be created by you. This means you are the one designing and implementing the algorithms. You must also be the one to write the initial report. Meaning the ideas and findings and summaries are yours. AI cannot be used for this. When in doubt, consult your TA or instructor.